

### • **Topic of interest**

In recent years, Smart fish farming has emerged as a promising approach to enhance the productivity, sustainability, and efficiency of aquaculture by integrating Internet of Things (IoT) and machine learning technologies. In integrated smart pond systems, the monitoring of water quality and fish farming recommendations relies heavily on the Internet of Things (IoT) and machine learning (ML). These systems utilize a variety of sensors to measure critical parameters such as temperature, turbidity, dissolved oxygen, and pH, providing real-time insights for maintaining optimal pond conditions. Collected data is sent to cloud-based repositories for storage and analysed using machine learning algorithms to recommend suitable fish species, optimize feeding schedules, and predict risks such as disease outbreaks. While these technologies show significant potential for improving productivity and sustainability in aquaculture, a major challenge lies in data integration. The data collected from diverse sensors and systems often exist in varying formats and resolutions, making it difficult to combine and process them effectively. This fragmented data hinders the development of unified decision-support systems, which are crucial for delivering accurate and actionable insights to farmers. Additionally, integrating large-scale datasets while maintaining data quality and consistency remains a technical hurdle. Inadequate integration may also lead to incomplete or biased analysis, limiting the reliability of fish species recommendations and predictive risk assessments. Addressing this issue requires robust data fusion techniques, standardized protocols for sensor outputs, and scalable systems capable of managing diverse data streams. By overcoming these challenges, smart pond systems can provide farmers with seamless, reliable, and data-driven solutions, ultimately advancing sustainable fish farming practices.

### • **Problem of Statement**

Small to medium-scale fish farmers face challenges in achieving accurate fish species recommendations and water quality monitoring due to class imbalance in machine learning models, sensor data variability, and system scalability issues. The existing systems often fail to handle imbalanced datasets, leading to inaccurate predictions, while sensor inconsistencies hinder real-time water quality monitoring. Additionally, many current solutions lack scalability for larger ponds, making them unsuitable for wide-scale adoption, particularly in rural or resource-limited environments. Islam *et al.* [14]: "*Fish Survival Prediction in an Aquatic Environment Using Random Forest Model*" – Highlights the limitations of fish species prediction models without preprocessing techniques like feature scaling and data

balancing. Pasika *et al.* [8]: "Smart Water Quality Monitoring System with Cost-Effective Using IoT" – Discusses real-time water quality monitoring but lacks automated decision support systems. Niswar *et al.* [10]: "IoT-Based Water Quality Monitoring System for Soft-Shell Crab Farming" – Demonstrates IoT implementation for aquaculture but lacks scalability and intelligent features. Islam *et al.* [16]: "An IoT Framework for Real-Time Aquatic Environment Monitoring Using Sensors" – Focuses on sensor-based monitoring but identifies gaps in decision support for fish farming. Fish farmers, particularly in small to medium-scale operations, struggle with implementing effective integrated smart pond water quality monitoring systems that provide accurate fish species recommendations. Current systems often suffer from class imbalance in datasets, leading to poor machine learning model performance and unreliable predictions. Moreover, real-time monitoring is hindered by sensor variability and limited system scalability, making it challenging for farmers in rural or resource-constrained areas to adopt these technologies. This study aims to develop a scalable IoT-enabled smart aquabot system that addresses these challenges by improving sensor accuracy, balancing machine learning datasets using advanced techniques like SMOTE, and delivering precise fish species recommendations. By enhancing these systems, fish farmers can make informed, data-driven decisions, increasing productivity, reducing operational costs, and promoting sustainable aquaculture practices. This problem statement clearly identifies who (fish farmers), what (issues with class imbalance, scalability, and sensor variability), when (during real-time monitoring and fish recommendation), where (small to medium-scale rural fish farms), why (to improve productivity and sustainability), and how (through IoT integration, data balancing, and machine learning enhancements), leading directly to the study's primary research question: How can an integrated IoT-based system optimize fish species recommendations and real-time water quality monitoring for sustainable fish farming?

- **Primary research question**

How can the recommendation of fish species for sustainable aquaculture be enhanced by the combination of machine learning algorithms and Internet of Things-based water quality monitoring?

- **Two specific aims/objective.**

1. To look into how well machine learning systems forecast appropriate fish species using data on water quality in real time. To achieve this goal, various machine learning models including Random Forest, SVM, and ensemble models will be trained and evaluated in order to determine

the optimal strategy for recommending fish species based on variables like pH, temperature, and turbidity.

2. To look into how IoT-based real-time water quality monitoring devices might enhance sustainability and decision-making in fish farming. The assessment of sensor integration for continuous data collecting, the difficulties of real-time data transmission, and the ways in which this data might support more sustainable and effective fish farming methods—especially in off-grid sites that are powered by solar energy—will be the main objectives of this aim.

#### • **Hypothesis**

The integration of IoT-based real-time water quality monitoring and machine learning for fish species recommendation will significantly improve the productivity and sustainability of fish farming by providing accurate, data-driven decisions for water quality management and species selection. This hypothesis reflects the expectation that the proposed system will enhance fish farming efficiency through real-time monitoring and automated decision support for selecting suitable fish species based on water quality.

#### • **Literature Review.**

The integration of emerging technologies such as the Internet of Things (IoT), machine learning (ML), and robotics has revolutionized various sectors, including agriculture and aquaculture. In the context of smart fish farming, the application of IoT for real-time water quality monitoring and ML for fish species recommendations has gained significant attention in recent years. Several studies have investigated the benefits and challenges of implementing these technologies in fish farming, with a focus on improving productivity, minimizing costs, and promoting sustainability. This literature review explores various research articles related to IoT-based water quality monitoring systems, fish species recommendations, and the integration of machine learning and renewable energy in smart aquaculture.

### **IoT-based Water Quality Monitoring in Fish Farming**

Water quality is a crucial factor in fish farming, as it directly influences fish health, growth, and survival rates. Monitoring water quality parameters such as temperature, pH, turbidity, dissolved oxygen (DO), and salinity is essential for maintaining optimal conditions for fish. However, traditional methods of water monitoring are often labour-intensive and time-consuming, making them less effective in ensuring real-time responses to sudden changes in water quality. Raju et al. [4] developed an IoT-based water quality monitoring system that

collects data from multiple sensors and transmits it to a cloud database for analysis. The system uses a Raspberry Pi microcontroller with an in-built Wi-Fi module to serve as the gateway for communication. The data is processed and sent to the farmer's device via a mobile application if water quality parameters fall outside the optimal ranges. While the system successfully monitors water quality, it does not provide any recommendation for selecting fish species based on water quality indicators, a key feature in enhancing the productivity of fish farming. In another study, Cordova-Rozas et al. [5] proposed a cloud-based water monitoring system comprising five phases: data extraction, cloud platform, database storage, reports, and predictions. Their system uses sensors to gather data, which is then transferred to a cloud platform for storage and processing. The recorded data are shown on dashboard reports, and predictions are made based on historical data to forecast future water quality parameters. Although the system provides valuable water quality data and predictions, it lacks an intelligent recommendation feature for fish species, which could have a significant impact on farming efficiency.

### **Machine Learning for Fish Species Recommendations**

The recommendation of appropriate fish species based on water quality is an important aspect of smart fish farming. The use of machine learning algorithms for analysing water quality data and recommending suitable fish species can help farmers make informed decisions and optimize their farming practices. Islam et al. [3,14] developed a model to predict suitable fish species based on water quality parameters such as pH, temperature, and turbidity. They applied machine learning algorithms, including Random Forest (RF), k-Nearest Neighbours (KNN), and J48, to analyse real-time pond data and achieved 88.48% accuracy with the RF model. This study demonstrates the potential of machine learning in making fish species recommendations based on water quality, but it primarily focuses on the accuracy of the models rather than the integration of these models into a fully automated system for real-time decision-making.

### **IoT and Renewable Energy Integration in Smart Aquaculture**

Sustainability is a key concern in smart aquaculture, and the integration of renewable energy sources, such as solar power, can play a significant role in reducing the carbon footprint of fish farming operations. Solar-powered systems are particularly attractive in remote areas, where access to electricity may be limited. Nagayo et al. [7] presented a solar-powered aquaponics system that incorporates a water-recirculating system for plant and fish growth, alongside a control system for monitoring and regulating water temperature. While the system provides a

multidisciplinary approach to aquaponics, it lacks an intelligent decision support system for choosing fish species based on water quality parameters. The integration of solar power with water monitoring could improve the sustainability and cost-effectiveness of the system, but the absence of data-driven decision support highlights a research gap in integrating renewable energy with machine learning-based decision systems. Similarly, Murad et al. [12] developed a solar-powered aquaponics system for monitoring water quality parameters like pH and temperature. The system uses sensors and GSM to notify farmers if parameters deviate from the ideal range. While this approach benefits from solar power, it lacks automated decision-making capabilities for recommending specific fish species, which limits its effectiveness in optimizing farming practices.

### **Smart Pond Management Using Robotics**

The use of robotics in smart aquaculture is another emerging trend that aims to automate the monitoring of water quality and the collection of data from remote locations in the pond. Robotic systems can help reduce the labor required for monitoring and enhance the overall efficiency of fish farming operations. Gao et al. [2,6] proposed an IoT-based fish farming and tracking control system that uses an intelligent management module for continuous water quality monitoring. The system integrates various sensors to collect water quality parameters, which are then analysed for forecasting future conditions. While this approach focuses on intelligent management of water quality, it does not incorporate a recommendation system for selecting fish species. Additionally, the study does not mention the use of renewable energy or robotics to automate the monitoring process.

### **Research Gaps and Future Directions**

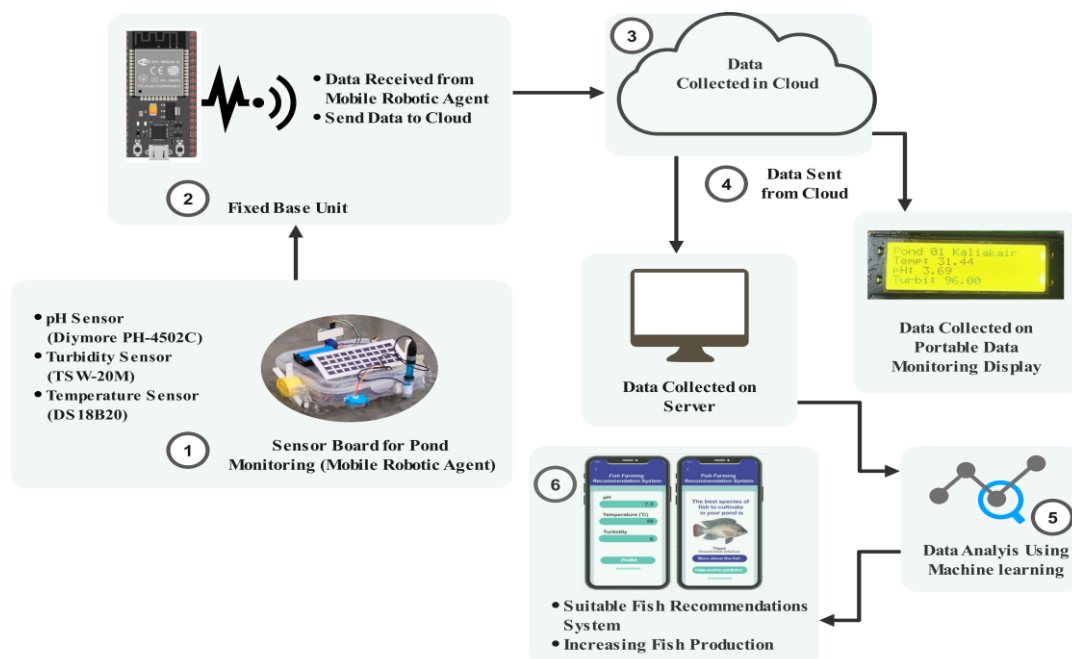
While numerous studies have explored the application of IoT and machine learning in fish farming, several gaps remain in the literature. First, the integration of intelligent decision support systems, particularly for recommending fish species based on real-time water quality data, is limited. Most existing systems [4, 5, 9-13] lack automated decision-making capabilities, which hinders the ability of farmers to make timely, data-driven decisions. Second, while many studies incorporate IoT-based water quality monitoring systems, there is a need for systems that combine IoT, machine learning, and robotics to provide a comprehensive solution for smart aquaculture. The integration of renewable energy sources, such as solar power, also remains underexplored, especially in the context of sustainable and self-sustaining systems for fish farming. Lastly, real-time predictive analytics for water quality is an area that requires further

research. Existing systems often focus on historical data analysis and predictions based on past conditions, but there is a need for systems that can predict and address potential water quality issues as they occur. This would enable farmers to take proactive measures, reducing the risk of fish mortality and improving overall farm productivity.

### • Design (Qualitative/Quantitative/Mixed)

### 3. Materials and Method:

Our Proposed Model Integrated System Architecture can be categorized into few sections from which we can address the overall architecture of the proposed model. (i) Mobile Robotic Agent: A wheeled robotic unit that gathers sensor data from the pond environment. (ii) Fixed Base Unit: Intermediary unit, this one receives the sensor data from the mobile robot and forwards it to the cloud; (iii) Cloud-Based Data Storage: A centralized repository for securely storing all collected sensor data. (iv) Remote Server + portable monitoring devices: it allows real-time monitoring and data collection to users. (v) Cloud-Based Data Processing: The server employs machine learning algorithms to analyse and derive insights from the collected data. (vi) Fish recommendations are made to farmers based on the analysed data to accommodate their fishes. A simple schematic diagram (Figure 1) represents the global system and the main functional components and connections. The flow of information and control between subsystems is clearly annotated in the diagram. Each of the subsystems is discussed in detail in the subsequent sections.



**Figure 1.** Integrated system architecture for the proposed model.

## Water Quality Parameters

This study focuses on three key parameters for monitoring water quality: pH, temperature and turbidity. Table 1 [1] provides us standard values for these parameters, which are essential for assessing the suitability of water for fish farming and identifying the most appropriate fish species for specific water conditions. Deviations from these standard values can significantly impact fish production and pose risks to the overall health of the aquaculture system. [1]

| Water Quality Parameter          | Value                 |
|----------------------------------|-----------------------|
| Temperature                      | 25°C –32 °C or >20 °C |
| pH                               | 6.5–8.5               |
| Turbidity                        | 30–80 cm              |
| Dissolved Oxygen (DO)            | >5 mg/L               |
| Biochemical Organic Demand (BOD) | <5 mg/L               |
| Total Dissolved Solids (TDS)     | 400 mg/L              |
| Chemical Oxygen Demand (COD)     | 20–30 mg/L            |
| Total Suspended Solids (TSS)     | <80 mg/L              |
| Electrical Conductivity (EC)     | 150–50 micros/cm      |
| Hardness                         | >15 mg/L              |
| Alkalinity                       | 50–300 mg/L           |
| Nitrite NO <sub>2</sub>          | <0.2                  |
| Nitrate NO <sub>3</sub>          | 0–100                 |
| Total Ammonia Nitrogen (TAN)     | 0–0.2                 |

**Table 1.** Reference values of each water quality parameter.

Temperature is the primary parameter we considered, as it directly influences other factors like dissolved oxygen (DO), conductivity, and salinity. Typically, these parameters are closely correlated with temperature. Generally, the temperature of the water is identical to the body temperature of fish and it increases or decreases proportionately [15]. A sudden temperature shift, even by 5 degrees Celsius, can cause stress or even lead to fish mortality. Cold water during winter often results in significant fish deaths. The second parameter, pH, plays a crucial role in aquaculture by regulating the acidity or alkalinity of soil and water. Measured on a scale from 0 to 14, prolonged high or low pH levels can hinder fish reproduction. Optimal pH levels for warm water range from 6.5 to 9, while a pH of 4 indicates highly acidic conditions that can be fatal for fish. Similarly, pH levels above 11 represent highly alkaline water, which also leads to fish mortality. Fish growth slows within a pH range of 5 to 6.5 or 9 to 10, and reproduction

ceases entirely between 4 and 5. Maintaining proper pH levels is essential as it helps mitigate harmful substances like hydrogen sulphide (H<sub>2</sub>S) and ammonia. Turbidity, a critical indicator of water quality, was examined in this study. A turbidity sensor helps measure the amount of debris in the water, as increased waste or particles can threaten fish survival. Turbidity levels offer a clear indication of water pollution, with higher levels negatively affecting aquatic life by limiting light penetration, damaging habitats, and disrupting photosynthesis. Additionally, elevated turbidity fosters conditions favourable for harmful bacteria and metals. Contaminants like algae, clay particles, plankton, fine organic and inorganic materials, and silt contribute to water pollution. Ensuring optimal water quality is essential for healthy fish growth and the prevention of diseases in aquaculture. However, environmental factors, such as weather, can greatly influence parameters like pH, temperature, and turbidity. Heavy rainfall may introduce acidic runoff from nearby land, lowering pH levels and stressing fish. On the other hand, algal blooms during warm weather can increase pH, affecting oxygen availability. Sudden temperature drops caused by storms can shock fish, while extended heatwaves reduce dissolved oxygen, leading to stressful conditions. Additionally, heavy rain can trigger erosion, causing sediment runoff that clouds the water, reduces light penetration, disrupts photosynthesis, and hampers fish feeding. Real-time monitoring of these parameters—temperature, pH, and turbidity—helps minimize fish stress, optimize resource utilization, and promote sustainable, profitable fish farming.

#### *Mobile Robotic Agent Design Subsystem*

We developed a lightweight, Bluetooth-controlled robot capable of autonomous navigation and data collection across a pond. The robot is equipped with an SG90 servo motor, a DC motor, an HC-05 Bluetooth module, an Arduino Uno, wheels, and an HCSR04 ultrasonic sensor. It uses obstacle detection to facilitate smooth navigation. Figure 2 illustrates the flowchart of the mobile robotic agent.



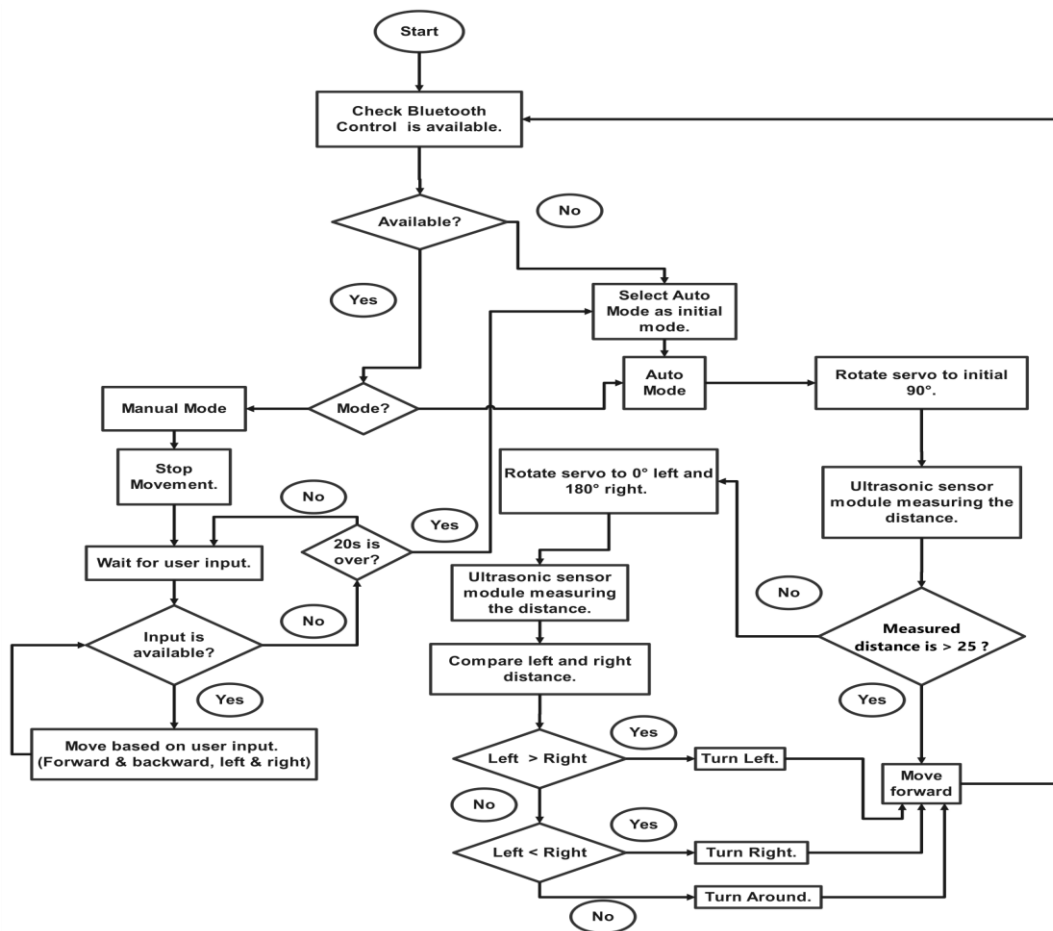


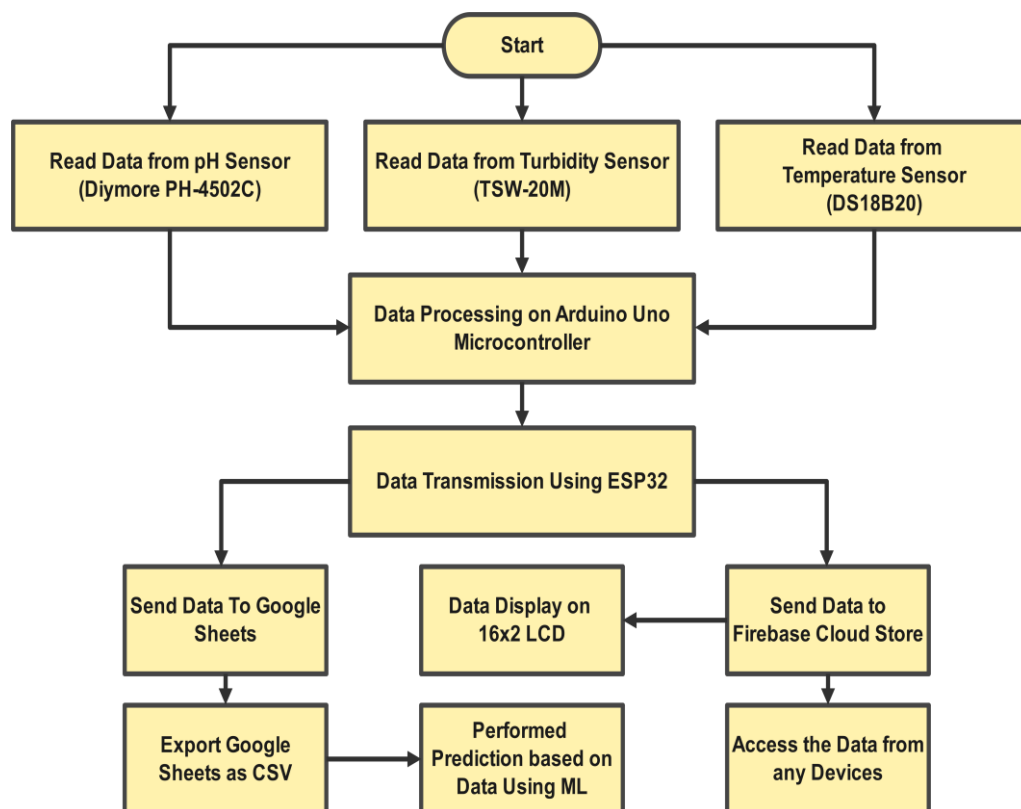
Figure 2. Flowchart of mobile robotic agent.

The robot operates by emitting ultrasonic waves through its trigger pin, which then reflect off nearby objects and are captured by the echo pin. This ultrasonic sensor has a detection range of 2 cm to 400 cm, with an accuracy of 3 mm, and can identify obstacles within a 30-degree angle. A servo motor is used to rotate the ultrasonic sensor, allowing it to scan for obstacles across a range of angles, with the motor shaft rotating between 0 and 180 degrees. To move the robot, a DC motor controlled by an L298D motor driver is employed. The motor driver acts as a current amplifier, enabling the DC motors to operate in either direction. The robot can be manually controlled via the "Pathfinder" smartphone application, which communicates with the robot using the HC-05 Bluetooth module. Users can guide the robot forward, backward, left, or right. In automatic mode, the ultrasonic sensor measures the distance to obstacles and determines whether to move forward, backward, or turn left or right. If an obstacle is detected within 25 cm, the servo motor rotates from 0 degrees to the left to 180 degrees to the right, selecting the direction with the most clearance. This allows the robot to continuously move forward. The robot is powered by solar energy, with a battery connected to store the energy required for operation. The solar panel automatically adjusts its position to face the sunlight, using a light-dependent resistor (LDR) and a servo motor to sense the sun's angle. Additionally,

the robot can measure water quality parameters from any location in a pond, leveraging its intelligent obstacle-avoidance capabilities.

#### *Data Acquisition, Transmission, and Monitoring Subsystem*

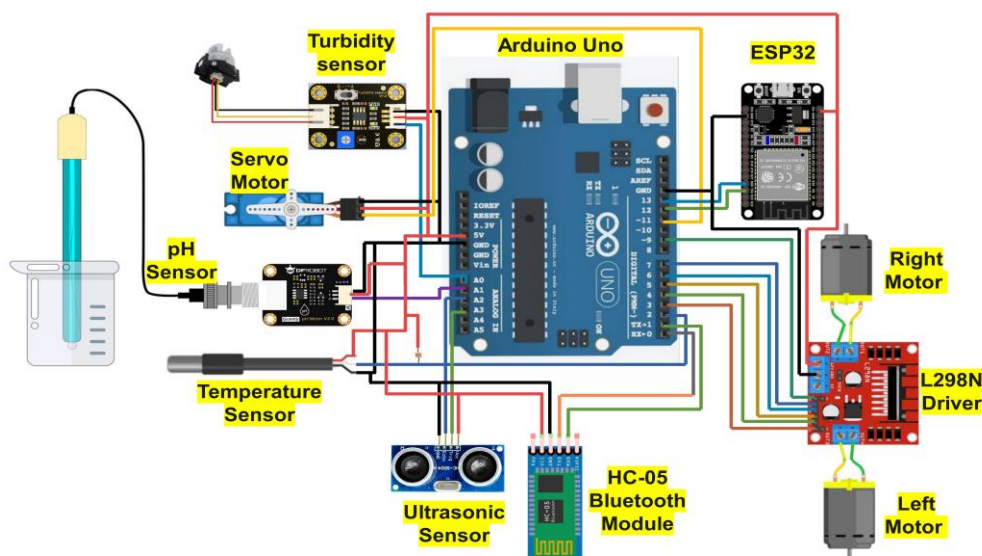
Our system incorporates a pH sensor (DiyMore PH-4502C), a turbidity sensor (TSW-20M), and a temperature sensor (DS18B20), along with an Arduino Uno microcontroller and an ESP32 liquid crystal display. These components work together to collect, process, transmit, and monitor data. Figure 3 illustrates the sequence of operations within this subsystem.



**Figure 3.** Flowchart of the proposed model.

The flowchart begins with the system start up, which involves powering on the robot and sensors and configuring the microcontroller to read sensor data. In the second stage, the sensors attached to the robot monitor water quality parameters, including pH, turbidity, and temperature. A waterproof DS18B20 temperature sensor measures water temperature, while the TSW-20M turbidity sensor assesses water clarity. The pH-4502C sensor records the hydrogen potential, with values ranging from 0 to 14. In the third stage, the ATmega328P microcontroller processes the sensor data, converting raw measurements into a format suitable for wireless transmission. The fourth stage establishes a wireless connection between the microcontroller and the ESP32 module. The ESP32 handles transmitting the processed data to

cloud-based platforms such as Google Sheets and Firebase for analysis and visualization. Additionally, the ESP32 processes the incoming data before delivery. The sixth stage focuses on analysing the cloud-stored data. Google Sheets serves as a real-time database where sensor readings are automatically updated each time the robot sends new data, enabling ongoing data analysis. Firebase facilitates real-time data visualization and analysis, while machine learning algorithms applied to Google Sheets data provide recommendations for optimizing fish cultivation practices. The final step involves user access to the data. Users can retrieve the stored information from the cloud platform through a web browser or mobile device, regardless of location. In addition to cloud-based monitoring, the system includes a portable remote monitoring device for real-time water quality measurements. This wireless, rechargeable device can be easily relocated to different pond areas. Connected to an ESP32 microcontroller, the device reads data from the Firebase cloud database and displays pH, temperature, and turbidity values on an I2C display. This portable solution enables farmers to check water quality across various pond locations conveniently without needing to access the cloud dashboard. It enhances the flexibility and usability of the monitoring system, allowing farmers to make informed decisions about fish cultivation. Figure 4 illustrates the system's circuit diagram. Arduino IDE version 2.3.2 was used to program the microcontroller, offering an intuitive interface for writing, editing, debugging, and uploading code to Arduino-compatible boards. The programming language utilized for this project was C++ or embedded C, with all coding tasks completed using the Arduino IDE.



**Figure 4.** Circuit diagram of the proposed system.

## • **Research Variables**

### **Independent Variables**

These are the variables that influence or affect the outcome. In the study, the independent variables are:

Water Temperature: The temperature of the pond water that affects fish survival and growth.

pH Level: The acidity or alkalinity of the water, which influences fish health.

Turbidity: The clarity of the water, which affects fish feeding and overall health.

Solar Power Utilization: This variable is concerned with whether the system utilizes solar power as a sustainable energy source to drive the IoT devices, sensors, and robotic components for water quality monitoring.

### **Dependent Variable**

This is the variable that depends on or is influenced by the independent variables. In your study, the dependent variable is:

Fish Species Recommendation – The type of fish suggested based on the water quality parameters (temperature, pH, and turbidity).

Fish Growth Rate: The rate at which fish grow in the monitored pond is an indirect dependent variable influenced by water quality and the appropriateness of the selected fish species.

Water Quality Deviation: The deviation from the ideal or optimal water quality parameters is another dependent variable.

## • **Scientific Method (Diagram)**

### *Fish Recommender System*

In this phase, we trained various machine learning models after collecting and pre-processing the dataset for advanced analysis. The conceptual diagram of the whole procedure is demonstrated in Figure 5.

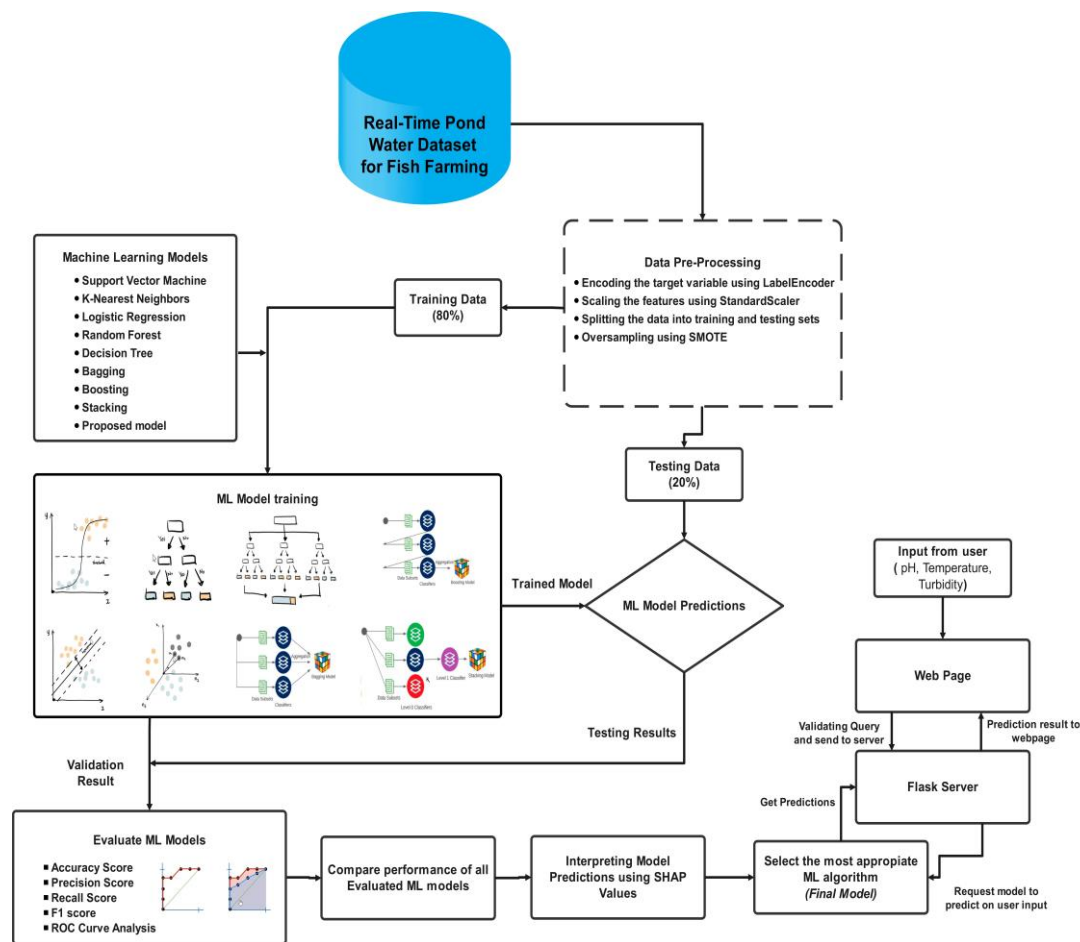


Figure 5. Conceptual diagram of ML model.

## • Citation and/or link to the article

## References

1. Ahmed,M.; Rahaman,M.O.; Rahman,M.; Abul Kashem,M. Analyzing the Quality of Water and Predicting the Suitability for Fish Farming based on IoT in the Context of Bangladesh. In Proceedings of the 2019 International Conference on Sustainable Technologies for Industry 4.0 (STI), Dhaka, Bangladesh, 24–25 December 2019; pp. 1–5. [CrossRef]
2. Zulkifli, C.Z.; Garfan, S.; Talal, M.; Alamoodi, A.H.; Alamleh, A.; Ahmaro, I.Y.Y.; Sulaiman, S.; Ibrahim, A.B.; Zaidan, B.B.; Ismail, A.R.; et al. IoT-Based Water Monitoring Systems: A Systematic Review. *Water* **2022**, *14*, 3621. [CrossRef]
3. Lu, H.Y.; Cheng, C.Y.; Cheng, S.C.; Cheng, Y.H.; Lo, W.C.; Jiang, W.L.; Nan, F.H.; Chang, S.H.; Ubina, N.A. A Low-Cost AI Buoy

System for Monitoring Water Quality at Offshore Aquaculture Cages. *Sensors* **2022**, 22, 4078. [CrossRef] [PubMed]

4. Raju, K.R.S.R.; Varma, G.H.K. Knowledge Based Real Time Monitoring System for Aquaculture Using IoT. In Proceedings of the 2017 IEEE 7th International Advance Computing Conference (IACC), Hyderabad, India, 5–7 January 2017; pp. 318–321.

[CrossRef]

5. Cordova-Rozas, M.; Aucapuri-Lecarnaque, J.; Shiguihara-Juarez, P. A Cloud Monitoring System for Aquaculture using IoT.

In Proceedings of the 2019 IEEE Sciences and Humanities International Research Conference (SHIRCON), Lima, Peru, 13–15

November 2019; pp. 1–4. [CrossRef]

6. Gao, G.; Xiao, K.; Chen, M. An intelligent IoT-based control and traceability system to forecast and maintain water quality in freshwater fish farms. *Comput. Electron. Agric.* **2019**, 166, 105013. [CrossRef]

7. Nagayo, A.M.; Mendoza, C.; Vega, E.; Al Izki, R.K.S.; Jamisola, R.S. An automated solar-powered aquaponics system towards agricultural sustainability in the Sultanate of Oman. In Proceedings of the 2017 IEEE International Conference on Smart Grid and Smart Cities (ICSGSC), Singapore, 23–26 July 2017; pp. 42–49. [CrossRef]

8. Pasika, S.; Gandla, S.T. Smart water quality monitoring system with cost-effective using IoT. *Heliyon* **2020**, 6, e04096. [CrossRef]

9. Huan, J.; Li, H.; Wu, F.; Cao, W. Design of water quality monitoring system for aquaculture ponds based on NB-IoT. *Aquac. Eng.* **2020**, 90, 102088. [CrossRef]

10. Niswar, M.; Wainalang, S.; Ilham, A.A.; Zainuddin, Z.; Fujaya, Y.; Muslimin, Z.; Paundu, A.; Kashihara, S.; Fall, D. IoT-based

Water Quality Monitoring System for Soft-Shell Crab Farming. In Proceedings of the 2018 IEEE International Conference on

Internet of Things and Intelligence System (IOTAIS), Bali, Indonesia, 1–3 November 2018; pp. 6–9. [CrossRef]

11. Pikulins, D.; Gotlaufs, R.; Solovjova, T.; Aboltins, A.; Grizans, J.; Migla, S. On the Development of Long-Range Water Quality

Monitoring System for Outdoor Aquaculture Objects. *Electr. Control Commun. Eng.* **2022**, *18*, 37–42. [CrossRef]

12. Murad, S.A.Z.; Harun, A.; Mohyar, S.N.; Sapawi, R.; Ten, S.Y. Design of aquaponics water monitoring system using Arduino

microcontroller. *AIP Conf. Proc.* **2017**, *1885*, 020248. [CrossRef]

13. Abinaya, T.; Ishwarya, J.; Maheswari, M. A Novel Methodology for Monitoring and Controlling of Water Quality in Aquaculture

using Internet of Things (IoT). In Proceedings of the 2019 International Conference on Computer Communication and Informatics

(ICCCI), Coimbatore, Tamil Nadu, India, 23–25 January 2019; pp. 1–4. [CrossRef]

14. Islam, M.M.; Kashem, M.A.; Uddin, J. Fish survival prediction in an aquatic environment using random forest model. *IAES Int. J.*

*Artif. Intell. IJ-AI* **2021**, *10*, 614. [CrossRef]

15. Amir, F.; Muchlisin, Z.; Nur, F.; Fadli, N.; Siti-Azizah, M.; Wilkes, M.; Tang, U.; Hasan, B.; Batubara, A.; Kocabas, F.; et al. Effect of

increasing water temperature on the physiology and gill histology of Barramundi, *Lates calcarifer* (Pisces, Perciformes) fingerlings.

*Int. Aquat. Res.* **2022**, *14*, 263–273. [CrossRef]

16. Islam, M.M.; Kashem, M.A.; Uddin, J. An internet of things framework for real-time aquatic environment monitoring using an

Arduino and sensors. *Int. J. Electr. Comput. Eng. IJECE* **2022**, *12*, 826–833. [CrossRef]